Analyzing Movement Through Intra-Personal Synchronization

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Introduction

Movement-based disciplines such as Tai Chi, are associated with lower levels of anxiety/depression. However, this relationship may be influenced by other factors like social engagement [1]. One potential approach to address this uncertainty could be quantifying movement learning. Traditional methods such as phase-portraits are difficult to apply to statistical evaluations and may misinterpret coordination due to frequency differences between limb segments [2]. This study uses computational synchronization metrics to quantify movement learning. The objective is to assess whether coordination improves after training and if these improvements correlate with mental health benefits.

Methods

Twenty-three volunteers (14 females, 9 males; aged 18–26) participated in the study. Recordings were made using eight infrared cameras and two force-plates. Helen Hayes protocol was used [3]. Participants stood on force plates and followed an instructor performing the exercise called Chan Si Gong. Scaled OpenSim models were used, employing inverse kinematics to extract joint angles. The following signals were analysed: left/right hip internal–external rotation, hip flexion, hip abduction, knee flexion–extension, pelvis rotation, lumbar rotation, and vertical ground reaction forces (left/right). The following synchronization metrics were applied: temporal mean phase-coherence, and re-normalized-mean-resultant-length (T), both utilizing Hilbert-derived phases [4, 5]. Figure 1 illustrates how T responds to coordination among four representative signals (joint angles and scaled ground reaction force (GRF)).

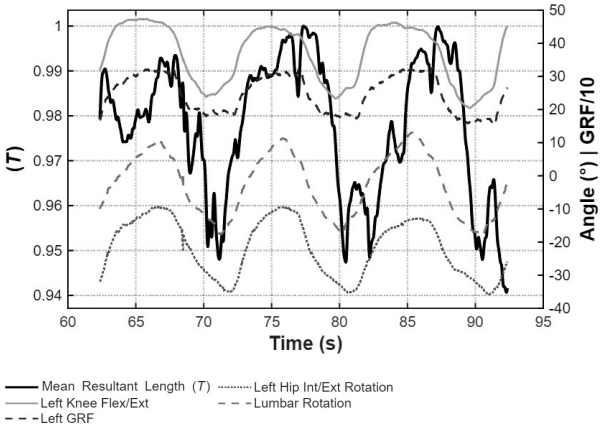


Figure 1. The level of synchrony between the signals at each time point affects T.

Additionally, psychological questionnaires were administered. The procedure was repeated after 4 weeks of Tai Chi–inspired training (3 sessions/week). Participants were divided by session attendance into Group Y (≥8 sessions) and Group N (<8 sessions).

Results

Data from two participants were excluded from the coordination analysis due to missing recordings and outlier detection. Results showed no significant coordination enhancements from pre-to-post training ($p=0.833$) and no significant changes in anxiety ($p=0.231$), a non-significant trend toward improved mindfulness ($p=0.041$; Bonferroni-adjusted $\alpha=0.017$), but significant increase in self-esteem ($p=0.011$). Table 1 shows the multivariate test results (mixed-design MANOVA) for the movement coordination. There was no statistical difference found between attendance and changes in coordination from pre- to post training.

Effect	Pillai's Trace	F(df1, df2)	p	Partial η^2	Observed Power
Attendance Group	0.252	F(2,18) = 3.040	0.073	0.252	0.515
Time	0.020	F(2,18) = 0.184	0.833	0.020	0.074
Time × Attendance Group	0.046	F(2,18) = 0.434	0.654	0.046	0.110

Table 1. Multivariate test results from the movement coordination MANOVA (n=21).

Conclusions

Intrapersonal coordination showed no significant improvement at the group level. Self-esteem increased as previously noted in literature [6]. As the coordination did not exhibit improvements, benefits in mental health observed by other short-duration studies may be unrelated to the learning of Tai Chi. The technique was useful for evaluating intra-personal synchronization.

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